

More On Inequalities

Most inequalities we have seen so far (even with rational expressions) have been "linear", possibly with absolute values.

What if we consider inequalities on rational expressions of higher degree?

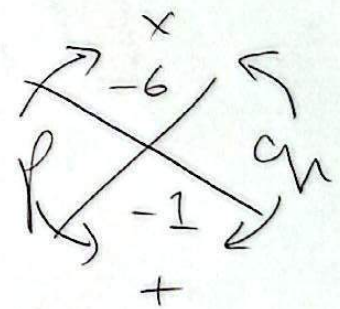
Let's start w/ a polynomial example

e.g. $2x^2 - x < 3$

$$\Rightarrow 2x^2 - x - 3 < 0$$

$$a = 2, b = -1, c = -3$$

$$a \cdot c = -6$$



$$p = -3, q = 2$$

$$\Rightarrow 2x^2 - 3x + 2x - 3 < 0$$

$$\Rightarrow x(2x-3) + 1(2x-3) < 0$$

$$\Rightarrow (x+1)(2x-3) < 0$$

We want $(x+1)(2x-3)$ to be negative

Step 1: Find zeroes of all factors

$$x = -1, x = \frac{3}{2}$$

$x = -2$ ← $x = 0$ → $x = 2$

	$(-\infty, -1)$	$(-1, \frac{3}{2})$	$(\frac{3}{2}, \infty)$
$x+1$	-	+	+
$2x-3$	-	-	+
$(x+1)(2x-3)$	+	-	+

$(-1, \frac{3}{2})$

e.g. $2x^2 - x \leq 3$?

$$2x^2 - x - 3 \leq 0$$

$$\Rightarrow (x+1)(2x-3) \leq 0$$

We want $(x+1)(2x-3)$ to be negative or 0

$$(-1, \frac{3}{2}) \cup \{-1\} \cup \{\frac{3}{2}\} = [-1, \frac{3}{2}]$$

e.g.

$$\frac{(2x+1)^2 (x-1)}{x(x^2-1)} > 0$$

$$\swarrow \quad \searrow$$
$$x^2-1 = (x-1)(x+1)$$

$$\frac{(2x+1)^2 (x-1)}{x(x-1)(x+1)} > 0$$

Diagram showing the cancellation of $(x-1)$ from the numerator and denominator, with arrows pointing to the resulting factors and their corresponding critical values:

- $x = -1/2$ (from $2x+1=0$)
- $x = 0$ (from $x=0$)
- $x = 1$ (from $x-1=0$)
- $x = -1$ (from $x+1=0$)

domain

$$\begin{aligned} x &\neq 0 \\ x &\neq 1 \\ x &\neq -1 \end{aligned}$$

As long as $x \neq 1$, we can cancel

$$\frac{(2x+1)^2}{x(x+1)} > 0$$

	$(-\infty, -1)$	$(-1, -\frac{1}{2})$	$(-\frac{1}{2}, 0)$	$(0, 1)$	$(1, \infty)$
$(2x+1)^2$	+	+	+	+	+
x	-	-	+	+	+
$x+1$	-	+	+	+	+
$\frac{(2x+1)^2}{x(x+1)}$	+	-	+	+	+

$$(-\infty, -1) \cup (-\frac{1}{2}, 0) \cup (0, 1) \cup (1, \infty)$$

e.g. $\frac{(2x+1)^2(x-1)}{x(x^2-1)} \geq 0$?

Where is numerator 0?
(denominator not 0)

$x = -\frac{1}{2}$ → only valid root

$$(-\infty, -1) \cup [-\frac{1}{2}, 0) \cup (0, 1) \cup (1, \infty)$$